A BOUNDED ABOVE COMPLEX OF PSEUDO-COHERENT MODULES IS PSEUDO-COHERENT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Cone lemma for pseudo-coherence in a distinguished triangle. Lemma \ref{lemma-cone-pseudo-coherent}, cited at proof lines 16245-16246. Performs the induction step along the truncation triangles.
- Being m-pseudo-coherent for all m is equivalent to being pseudo-coherent. Lemma \ref{lemma-pseudo-coherent}, cited at proof lines 16248-16249. Derives the final statement about pseudo-coherent modules.

Source reference. Original lemma 064Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7882–7882. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 064R: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7875–7875. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 064U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7878–7878. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let R be a ring. Let $m \in \mathbf{Z}$. Let $K^\bullet$ be a bounded above complex of R-modules such that K^i is $(m-i)$-pseudo-coherent for all i. Then $K^\bullet$ is m-pseudo-coherent. In particular, if $K^\bullet$ is a bounded above complex of pseudo-coherent R-modules, then $K^\bullet$ is pseudo-coherent.*

Proof. We may replace $K^\bullet$ by $\sigma_{\geq m-1}K^\bullet$ (for example) and hence assume that $K^\bullet$ is bounded. Then the complex $K^\bullet$ is m-pseudo-coherent as each $K^i[-i]$ is m-pseudo-coherent by induction on the length of the complex: use Lemma 064R and the stupid truncations. For the final statement, it suffices to prove that $K^\bullet$ is m-pseudo-coherent for all $m \in \mathbf{Z}$, see Lemma 064U. This follows from the first part. □